Question Paper Code: 25900463

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2025

THIRD SEMESTER

Discipline Specific Core Course (DSC)
MG3DSCICH200 – FUNCTIONAL OPERATIONS IN CHEMICAL INDUSTRY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Instruction to the Examiners:

- This scheme is only a guideline for valuation. Any other attempt which is equally valid and correct may be given due marks.
- Exceptions are there in some cases and if the answer is complete full marks may be given even if the required number of valuation points could not be identified.
- Some of the answers given in the scheme are elaborate and this is to help the examiner in their valuation. Please do not expect all those points in the answers of the students.

Part A

Multiple choice questions

Answer all questions

Each question carries 1 marks

1. The most common type of tray used in distillation columns is **[E] / [CO1]**
   Answer c). Sieve tray
2. The mixing mechanism in a tumbler mixer is primarily: **[An] /[CO3]**
   Answer a). Convective mixing due to tumbling motion
3. Bernoulli's principle is based on the conservation of **[K] / [CO1]**
   Answer: b). Energy
4. Which method is also called flame fusion? **[K] / [CO2]**
   Correct Answer not in option
   Give full credit to all
5. Which of the following is NOT true for batch distillation? **[E] / [CO1]**
   Answer:d) Produces constant composition distillate
6. Nutsche filter is a: **[An] /[CO1]**
   Answer : b) Batch filtration technique using vacuum or pressure
7. In an MSMPR crystallizer, what does MSMPR stand for? **[K] / [CO2]**
   Answer:a) Mixed Suspension Mixed Product Removal
8. The function of the wiper blades in a wiped film evaporator is to: **[U] / [CO1]**
   Answer: b). Spread liquid into a thin turbulent film
9. The process of transferring mass from one phase to another is called **[U] / [CO1]**
   Answer : b). Mass transfer
10. Which of the following is NOT an advantage of forced circulation evaporators? **[E] / [CO1]**
    Answer: c). Short residence time

[1x10 = 10]

Part B
Short Answer Questions
Answer any 4 questions
Each question carries 3 marks

11. Why is steam distillation used for separating essential oils? [U] / [CO1]

Steam distillation is used because essential oils are typically heat-sensitive (thermolabile) and have high boiling points.

Lower Boiling Point, Prevents Decomposition, Easy Separation

Any 3 relevant points with explanation give full credits

12. Write a note on the solvents used for liquid-liquid extractions. [K] / [CO3]

Liquid-liquid extraction is a separation technique in which a solute is transferred from one liquid phase to another based on its relative solubility in the two immiscible liquids—usually an aqueous phase and an organic solvent.

Characteristics of an Ideal Solvent for Liquid-liquid extraction

High Selectivity, High Distribution Coefficient, Low Miscibility with the Other Phase, Low Density Difference, Low Viscosity, Chemical Inertness, Low Toxicity and Flammability,

Easy Recoverability and Reusability

Reasonable explanation with any 3 points give full credit

13. Explain how filter aids increase the rate of filtration. [E] / [CO1]

A filter aid is a hard incompressible material which is added to the slurry in a very finely divided form. The presence of filter aid increases the porosity of the cake, decreases its compressibility and reduces the resistance of the filter cake. Besides which can form a layer of second medium which protect the basic medium of the system. This is commonly referred to as precoat. Precoat prevents gelatinous solids from plugging the filter medium and give clear filtrate.

Proper explanation give full credit

14. **Mention two applications of rotary dryer. [E] / [CO1]**

1. Drying of Minerals and Ores
2. Drying of Agricultural and Food Products

Any two relevant applications with explanation. Each application 1.5 marks each

15. **Define mixing. Why is it important in chemical and pharmaceutical industries? [An] /[CO3]**

Mixing is a unit operation that aims to reduce non-uniformity or gradients in a system. Its primary goal is to combine two or more separate components to achieve a desired level of homogeneity. This can involve blending different liquids, dissolving solids in liquids, dispersing a gas in a liquid, or intermingling particulate solids.

In the Chemical Industry -Uniform Reaction Mixtures,Enhanced Heat and Mass Transfer,Consistent Product Quality,Catalyst Distribution,Process Safety etc

In the Pharmaceutical Industry-Uniform Drug Formulation,Improved Bioavailability,Quality Control, Stable Products, Efficient Coating and Granulation etc

Definition 1mark. Any reasonable in chemical industry-1 mark pharmaceutical -1 mark

16. **Explain the principle of the Bridgman technique in crystal growth. [U] / [CO2]**

The Bridgman technique is a method for growing large single crystals from a molten charge, based on controlled directional solidification.In the Bridgman method, the molten material is slowly cooled so that it solidifies gradually from one end to the other inside a container (crucible).A temperature gradient is maintained — one end of the furnace is hotter (above melting point) and the other is cooler (below melting point).As the crucible is slowly moved from the hot zone to the cool zone, the melt solidifies progressively, allowing the formation of a single large crystal along the length of the container, guided by the controlled temperature gradient.

Principle -1mark Explanation-2marks [3x4 = 12]

Part C
Short Essay Questions
Answer any 4 questions
Each question carries 7 marks

17. Explain the principle of crystallization in detail. Discuss how solubility, temperature, and pressure affect the process [U] / [CO2]

The core principle relies on achieving a state of supersaturation, where the concentration of the solute in the solution is higher than its maximum solubility at a given set of conditions. This unstable state provides the

thermodynamic driving force for the excess solute to precipitate out as pure crystals. The process occurs in two main stages:

1. Nucleation 2. Crystal Growth

Effect of Key Factors

Solubility-Impurities must remain soluble in the solvent even at lower temperatures, allowing the pure substance to crystallize out, effectively separating the two.

Temperature-For most solid solutes, increasing the temperature increases their solubility in a liquid solvent.-Nucleation and Growth Rates-Polymorphism

Pressure-Effect on Solids/Liquids- Changes in pressure have a minimal effect -Effect on Gases- According to Henry's Law, the solubility of a gas is directly proportional to its partial pressure above the liquid Crystal Properties- high pressure can affect the crystallization temperature of certain brines and can also influence the density and hardness of the resulting crystals.

Proper Explanation with each stages and principle -4marks factors-3 marks

18. **Give a detailed account of the uses of extraction in qualitative and quantitative analysis.** **[U] / [CO3]**

Uses in Qualitative Analysis-Isolation of Target Compounds, Selective Removal of Interferences, Preliminary Screening of natural products, Coupling with Identification Techniques

Uses in Quantitative Analysis-Extraction is vital for sample preparation to ensure accurate and precise measurements. Sample Cleanup, Enabling Measurement, Determining Yield and Purity

Environmental Monitoring and Food Safety, Biomedical Applications.

Any 7 relevant uses give full credits: Qualitative-4marks Quantitative-3 marks

19. **During Supercritical Fluid Extraction of hops for beer flavoring, researchers use ethanol as a co-solvent along with CO_2 .** **[U] / [CO3]**

a)What role does the co-solvent play? b)Why might CO_2 alone be insufficient?

a) The Role of the Co-Solvent (Ethanol)-In the SFE of hops, ethanol acts primarily as a modifier or entrainer,Increases Polarity-Enhances Interaction.

b) Why CO2 Alone Might Be Insufficient-In the context of hops, pure CO2 extraction might leave behind significant quantities of valuable, more polar compounds essential for a full-spectrum extract-Co-Solvent ensures a more efficient and comprehensive extraction process.

Relevant explanation give full credits: definition or explanation of SFE-2marks a)3marks b)2 marks

20. **Describe in detail the stages of crystallization: supersaturation, nucleation, and crystal growth, with suitable diagrams**. **[U] / [CO2]**

The stages of crystallization are supersaturation, nucleation-primary and secondary, and crystal growth-diffusion,surface integration and rate control, occurring in sequence to transform a dissolved solute into a highly ordered solid crystal structure.

Proper explanation of three stages -5marks: diagram-2 marks

21. **Discuss the importance of reflux ratio in rectification. How does it affect the efficiency, separation quality, and energy consumption of the distillation process?** **[An] /[CO1]**

Any relevant points related to rectification give full credits

22. **Write an essay on how simple distillation illustrates the principle of volatility differences.** **[U] / [CO1]**

Simple Distillation is based on the fundamental principle that different substances have different boiling points and volatilities. The process of simple distillation exploits these differences. When a liquid mixture is heated, the component with the **lower boiling point** (higher volatility) vaporizes first. This vapour, when condensed, forms a distillate that is richer in the more volatile component. The operation of simple distillation clearly demonstrates that the separation of components is possible only if there is a sufficient difference in their boiling points — usually at least 25–30°C.For example, when a mixture of acetone (b.p. 56°C) and water (b.p. 100°C) is distilled, acetone, being more volatile, vaporizes first and condenses as the distillate, leaving water behind.

Principle -2 marks Relevant explanation with any example-5marks

[7x4 = 28]